

Mark schemes

1.

Level of response	Response	Max 3 marks
3	A full and precise definition is given.	3 marks
2	The substantive content of the definition is correct, but there may be some imprecision or inaccuracy.	2 marks
1	Some fragmented points are made.	1 mark

Example of acceptable definition worth 3 marks:

- total revenue minus total cost.

Examples of a definition worth 2 marks:

- revenue minus cost (no total)
- the difference between total revenue and total cost (no direction)
- the reward to an entrepreneur.

Examples of a definition worth 1 mark:

- the difference between revenue and cost (no total or direction)
- money made by firms.

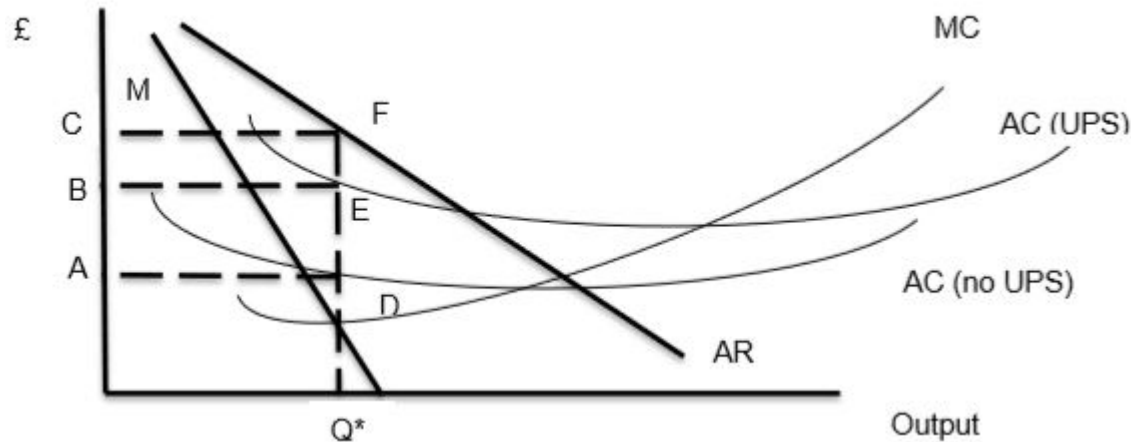
[3]



2.

Level of response	Response	Max 9 marks
3	• is well organised and develops one or more of the key issues that are relevant to the question • shows sound knowledge and understanding of relevant economic terminology, concepts and principles • includes good application of relevant economic principles and/or good use of data to support the response • includes well-focused analysis with a clear, logical chain of reasoning • includes a relevant diagram that will, at the top of this level, be accurate and used appropriately	7 – 9 marks
2	• includes one or more issues that are relevant to the question • shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles and/or data to the question • includes some reasonable analysis but it might not be adequately developed and may be confused in places • may include a relevant diagram	4 – 6 marks
1	• is very brief and/or lacks coherence • shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • demonstrates very limited ability to apply relevant economic principles and/or data to the question • may include some very limited analysis but the analysis lacks focus and/or becomes confused • may include a relevant diagram but the diagram is not used and/or is inaccurate in some respects	1 – 3 marks

Suggested Diagram:



- A monopoly diagram is required. The monopoly diagram could show two average cost curves, with one indicating the higher level of average costs for Royal Mail operating under the UPS obligations and the other (lower) one indicating the lower average costs if the UPS was not present.
- The diagram could show the different profits made under the two scenarios, either by shading, or by labelling as above (rectangle ACFD represents profits without UPS, whereas BCDE represents profits with UPS).

Relevant issues include

- definitions of monopoly, costs, profits
- explanation of what the UPS is (6 days a week delivery to all addresses in the UK)
- explain that only Royal Mail has the UPS obligation, so there is an unfair market, where competitors can “cherry pick” the most profitable geographic areas or types of business
- larger fleet of vehicles, greater number of staff, more mail depots are all required to provide the necessary scale for delivery to all addresses
- increased time and costs of delivering to rural areas
- explanation of how higher average costs tend to lead to lower profits

Note

Some candidates might also consider how the UPS obligations might affect Royal Mail's variable and hence marginal costs. If reasonable, such an approach should be rewarded.

Other approaches to the question and related diagrams should be rewarded.

**3.****Areas for discussion include:**

- explanation of privatisation and deregulation
- the arguments in favour of privatisation in general
 - efficiency
 - investment
 - greater responsiveness to customer needs
 - adoption of best practice (including from foreign firms)
 - building a shareholder culture
 - one-off impact on public finances
- the arguments against privatisation in general
 - higher prices due to monopoly/oligopoly power
 - reduced quality of service
 - reduced consistency of service
 - greater risk of externalities
 - difficulties in identifying the true value of state-owned assets at time of privatisation
 - costs incurred in the regulation of newly privatised industries
 - impact on employees and trade unions
- the arguments for and against introducing competition alongside privatisation
- impact on customer service quality, reliability and punctuality
- impact on customer price
- profitability of private operators
- impact on technological innovation and dynamic efficiency
- value for money for the taxpayer, perhaps with reference to the criticisms of the Government's handling of the sale of Royal Mail.
- impact on workers' job security, pay and conditions
- regulation of Royal Mail – the split model of competition in parcels but virtually a monopoly in letters allows for comparisons to be made about the different approaches
- reference to the fact that letter collection is more competitive than letter delivery, and the problems faced by Whistl in attempting to enter the latter market
- alternatives to nationalisation and privatisation: such as contracting out, subsidies
- the possibility of government failure, either in running a nationalised post service, or in organising the privatisation process
- market failure arguments.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

Level of response	Response	Max 25 marks
5	Sound, focused analysis and well-supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response - includes well-focused analysis with clear, logical chains of reasoning - includes supported evaluation throughout the response and in a final conclusion.	21 – 25 marks
4	Sound, focused analysis and some supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response - includes some well-focused analysis with clear, logical chains of reasoning - includes some reasonable, supported evaluation.	16 – 20 marks
3	Some reasonable analysis but generally unsupported evaluation that: - focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present - includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response - includes some reasonable analysis but which might not be adequately developed or becomes confused in places - includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgements but these are not well-supported by arguments and/or data.	11 – 15 marks
2	A fairly weak response with some understanding that: - includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - includes some limited application of relevant economic principles to the given context and/or data to the question - includes some limited analysis but it may lack focus and/or become confused - includes some evaluation which is weak and unsupported.	6 – 10 marks
1	A very weak response that: - includes little relevant knowledge and understanding of economic terminology, concepts and principles - includes application to the given context which, at best, is very weak - includes attempted analysis which is weak and unsupported.	1 – 5 marks

[25]

5.

C

[1]

6.

A

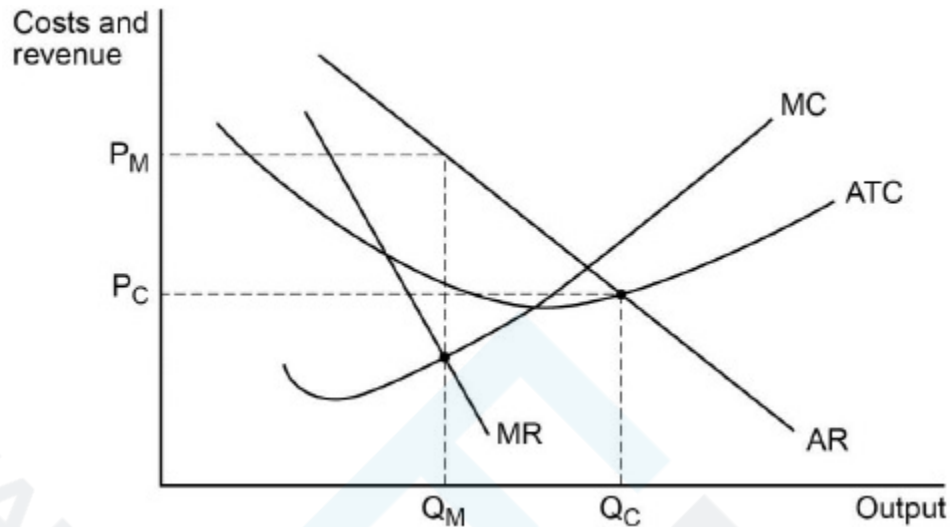
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7.

Level of response	Response	Max 9 marks
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2	- includes one or more issues that are relevant to the question - shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present - includes reasonable application of relevant economic principles and/or data to the question - includes some reasonable analysis but it might not be adequately developed and may be confused in places - may include a relevant diagram.	4 – 6 marks
1	- is very brief and/or lacks coherence - shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - demonstrates very limited ability to apply relevant economic principles and/or data to the question - may include some very limited analysis but the analysis lacks focus and/or becomes confused - may include a relevant diagram but the diagram is not used and/or is inaccurate in some respects.	1 – 3 marks

Suggested Diagram:

- a monopoly diagram is expected, showing a firm profit maximising initially, but subsequently reducing prices closer to (or at) normal profit. A monopolist facing potential entrants in a contestable market would fear entry and reduce price from P_M to P_C , with profits falling to just normal profits



- other reasonable diagrams are acceptable, eg monopolistic competition with the AR/MR shifting left because of entry.

Relevant issues include

- explanation of contestability and barriers to entry in the banking industry
- explanation of “new players”, including examples, such as Metro Bank
- why increased competition usually reduces prices and profits
- analysis of how supernormal profits can be driven towards zero by the presence (or even mere threat) of new entrants
- reference to increasing innovation and focus on customer service of new entrants
- explanation of how improving the customer switching service may reduce customer inertia
- reference to falling revenue per customer.

[9]

8.

D

[1]

**9.****Areas for discussion include:**

- definition of profit, normal profit, supernormal profit, firm
- the need for firms to make at least normal profit to remain in their present use, including short run vs long run requirements
- firms use of supernormal profits as a necessary inducement for risk-taking, source of investment funds, as a reward to the owners of the firm, as an indicator of the firm's health as a going concern and as a source of working capital
- the role of profit in the wider economy
 - as the incentive to establish and run any enterprise in a free market economy
 - investment in technology and innovation, eg public benefits of new drugs
 - as a source of income, especially for pensioners
 - as a signalling device for high growth industries, to promote the efficient allocation of resources within an economy
 - as a source of tax revenue.

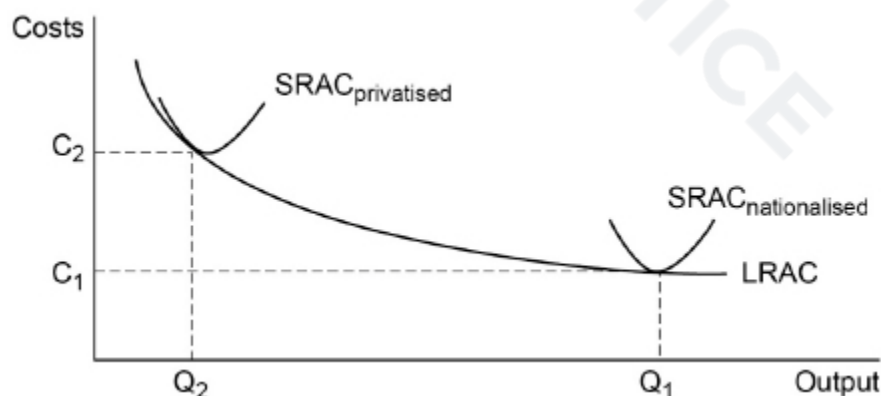
The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

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2	A reasonable response provides an answer that: • focuses on issues that are relevant to the question • shows satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places.	6 – 10 marks
1	A weak response provides an answer that: • has identified one or more relevant issues • has some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • has very limited application of relevant economic principles to the given context and/or data to the question • might have some limited analysis but it may lack focus and/or become confused.	1 – 5 marks

[15]

Level of response	Response	Max 9 marks
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A natural monopoly diagram is expected, showing constantly declining LRAC in the rail sector. In addition, two or more SRAC curves contained within the LRAC envelope could be used to illustrate the different cost profiles of the sector, though these are not essential. One SRAC to show where the opportunities for economies of scale aren't adequately exploited and the other showing where the industry is operating as a monopoly, exploiting more fully the potential economies of scale.





Relevant issues include:

- explain what is meant by long-run average costs
- explanation of the characteristics of natural monopoly, and why railways can be considered a natural monopoly (huge network, large capital costs, etc)
- explanation of minimum efficient scale relative to the size of the market
- how automation and technological progress boost labour productivity within industries and markets
- provide examples of economies of scale that might be available to firms operating in the railway sector
- reference to the fact that costs are 40% higher in the UK than the rest of Europe
- explanation of the higher long run average costs that are likely to flow from fragmenting the railways into numerous smaller parts
- explain the franchise model and how this encourages short-term priorities at the expense of long-run average cost reductions
- explain that the profit motive forces firms to cut costs in order to maximise profit – this could be via a reduction in X-inefficiency.

[9]

11.

Areas for discussion include:

- what is meant by 'firm'
- explanation of profit, drawing a distinction between normal and abnormal profit
- explanation of profit maximisation
- why profit is usually an important objective
 - reward to enterprise
 - source of finance for investment
 - signalling corporate health to shareholders
- how firms in perfectly competitive sectors have little choice but to pursue profit-maximisation but firms with monopoly power have greater flexibility
- how ownership of the firm affects behaviour and performance
- how the divorce of ownership from control and the principal-agent problem affect behaviour and performance
- revenue maximisation and sales maximisation as alternative objectives
- limitations to the feasibility of pursuing profit maximisation, for example the difficulty many firms face in calculating marginal revenue and marginal cost
- discussion of profit satisficing rather than maximising
- sales growth as a goal in itself, or as a short-term means to a long-term goal of profit maximisation
- non-financial objectives such as environmental goals and corporate social responsibility
- whether the assumption of profit-maximisation is any more or less valid than in the past
- examples of firms which have/have not pursued the aim of profit maximisation.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the student's response to the question.

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5	Sound, focused analysis and well-supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response - includes well-focused analysis with clear, logical chains of reasoning - includes supported evaluation throughout the response and in a final conclusion.	21 – 25 marks
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2	A fairly weak response with some understanding that: - includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - includes some limited application of relevant economic principles to the given context and/or data to the question - includes some limited analysis but it may lack focus and/or become confused - includes some evaluation which is weak and unsupported.	6 – 10 marks
1	A very weak response that: - includes little relevant knowledge and understanding of economic terminology, concepts and principles - includes application to the given context which is, at best, very weak - includes attempted analysis which is weak and unsupported.	1 – 5 marks

**12.****Areas for discussion include:**

- definitions of short run and long run
- definition of normal profit, awareness that it should be considered a cost and why it is required to remain in business in the long run
- explanation of profit maximisation
- explanation of what is meant by loss, in other words total costs exceeding total revenues
- explanation of why this is likely to be problematic in the short run (eg cash-flow problems) and long run
- analysis of breakeven point for a firm and the need for average revenue to cover average total costs at the profit-maximising point in the long run
- analysis of the need for average revenue to cover average variable costs at the profit-maximising point in the short run, to avoid the firm shutting down
- explanation of why a loss-making firm is able to carry on as long as it is covering its variable costs
- the availability of options for continuing if running operations at a loss, eg the existence of affordable credit or investment, or previously retained earnings
- analysis of barriers to exit such as sunk costs
- application to perfectly and/or imperfectly competitive markets
- impact of business cycles on a firm's profit – cycles may be caused by booms and recessions, or by seasons
- firms may consider long-run profits rather than short-run profits and continue despite making losses
- in the short run the firm may be investing heavily in capital, R & D or marketing, which may lead to short-run losses but long-run profits
- in the short run the firm may be pursuing strategies to increase market share, and gain monopoly power in the long run
- use of examples, including those mentioned in the quote, to illustrate why a firm may to continue to produce in the short run, despite making losses.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the student's response to the question.

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[15]

13.

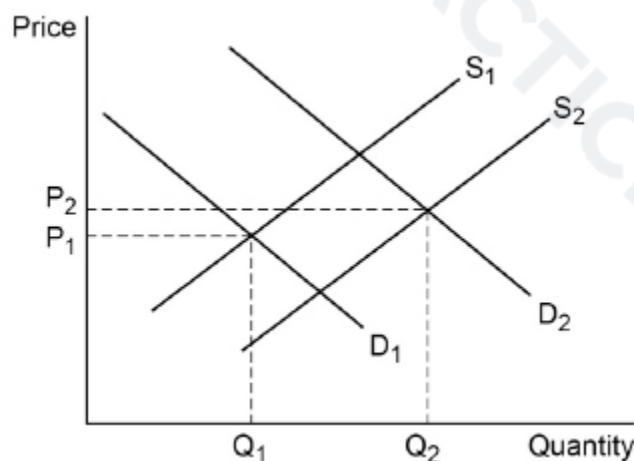
A or C

[1]

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Suggested diagram

A supply and demand diagram is expected, showing a shift right in the supply curve and a shift right in the demand curve, resulting in higher levels of output.





Two separate diagrams are acceptable.

Other relevant diagrams, such as cost curve diagrams, may be rewarded as part of a logical explanation.

Relevant issues include:

- explanation of cheaper transportation
- explanation of construction boom
- explanation of derived demand
- how cheaper transportation shifts the supply curve out
- how construction booms trigger greater demand for sand (via land reclamation in Singapore's case, or as an ingredient in cement and other building materials)
- how cheaper shipping (for example) can create new opportunities for trade between countries
- explanation of how cheaper shipping, by itself, would create excess supply at the original equilibrium price, thus resulting in lower prices and some additional demand
- explanation of how a construction boom, by itself, would create excess demand at the original equilibrium price, thus resulting in higher prices and some additional supply
- analysis of how the combination of demand and supply shifts may enable the sector to grow without prices rising
- reference to extracts including **Figure 2**, Chinese construction statistics, the export of Australian sand to Dubai and 'megacities'.

[9]

15.

B

[1]

16.

B

[1]

17.

Areas for discussion include:

- definitions of perfect competition, productive efficiency, allocative efficiency
- explanation of the assumptions of perfect competition
- explanation of how and why firms are forced to be price-takers in perfect competition
- the ability to earn supernormal profits is restricted. In the short run, perfectly competitive firms may make some supernormal profit, in the long run such profits will be eroded by new entrants joining the market or existing competitors expanding output. This contributes to allocative and productive efficiency
- explanation of how firms must produce output at the lowest point on the average cost curve. Any productive inefficiency will be punished by other more efficient firms selling at a lower price
- explanation of the condition for allocative efficiency ($P=MC$) and how and why this will be ensured in perfect competition.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the student's response to the question.

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[15]

**18.****Areas for discussion include:**

- definition of technological change, perfect competition
- discussion of the features of competitive markets in relation to technological change and market structure
- recognition that perfect competition is an unattainable ideal in many sectors, and is best thought of as one end of a spectrum of market structures to which sectors should (arguably) be guided
- in some sectors, the use of the internet and other technological innovation has hugely improved price transparency, thus minimising information asymmetry and getting closer to perfect information (eg price comparison websites, insurance brokers)
- in some sectors, firms have strengthened dominant positions by erecting barriers to entry such as massive economies of scale and advanced inventory systems (eg Amazon)
- the impact of artificial intelligence, robotics, and retention of personal data on markets, making markets less competitive, eg online customers now receiving personalised adverts
- the improvement of communications has led to improved information flows and made some markets more competitive, but has also led to the demise of other sectors, eg streaming replacing CDs and radio to an extent
- the impact of technology on reducing average costs and thus potentially reducing prices
- the impact of technology on the number of sellers in a market, eg possibility of starting a business with a laptop from a bedroom
- the impact of technological change on homogeneity of products and services
- discussion of productive, allocative and dynamic efficiency
- discussion of Schumpeterian creative destruction
- examples of markets that have become more competitive
- examples of markets that have become more concentrated
- overall assessment about whether technological change tends to bring industries closer to the market structure of perfect competition.

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**19.**Calculation: $\$2.6\text{bn} \times 1.8\% = \$46\,800\,000$

Correct answer is: \$47 million

Response:	Max 2 marks
For the correct answer, to nearest million, with \$ sign	2 marks
For the correct answer but without the \$ sign and/or not to nearest million OR For the correct method, but the wrong answer, to nearest million, with the \$ sign	1 mark

[2]

20.**Areas for discussion include:**

- explanation of pharmaceutical market structure
- discussion of consumer interests in this market: low and stable prices, safe and effective products, access to drugs (these are merit goods)
- discussion of the role the NHS plays in acting as a monopsonistic buyer of pharmaceuticals in the UK, whereas in the USA and elsewhere, there are many more buyers
- the oligopolistic nature of the generics market, with a small number of firms typically producing each drug, and the opportunity for collusion and price-fixing, especially as there is little or no branding of products
- the fact that monopoly power is legally granted to firms in order to encourage research and development into new drugs. The advantage of this is that new drugs are invented. The disadvantage is that it can take 20 years for lifesaving drugs to become more affordable
- the ability of pharmaceutical firms to exploit monopoly power and inelastic demand by massively increasing prices
- the importance of economies of scale
- the importance of dynamic efficiency and links to market structure
- the significance of productive and allocative efficiency
- in generic pharmaceutical markets, oligopoly power can lead to tacit or collusive price-setting
- very high levels of investment in research and development for new drugs and treatments, contributing to rising life expectancy and improved survival rates, for example cancer and HIV/AIDS
- discussion of possible policies at the national level which may impact on consumer interests, eg regulation by the CMA
- market failure arguments
- government failure arguments.

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[25]

21.

B

[1]

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Remember: AO4, ie evaluation, is not being assessed through this question.

Relevant issues include:

- meaning of forecast and the inevitable uncertainties involved when forecasting future sales
- identifying that the growth in the demand for aircraft is derived from the demand for both passenger and freight air transport
- explain the significance of the growth in the world economy and cyclical fluctuations in economic activity
- explain the significance of the growth in the middle-class population
- linking the above to the growth in tourism, the demand for air travel and hence aircraft
- the income elasticity of demand for air transport
- expected developments in the globalisation of the world economy
- the relative price of the company's aircraft
- explain the relevance of both price and cross elasticities of demand
- the effect of changes in exchange rates
- the rate of growth in different market segments and how they link to the type of aircraft the company sells
- the effect of new entrants into the market, eg COMAC
- the extent to which the airlines need to replace ageing aircraft
- changes in technology and the impact on the cost of operating aircraft
- changes in technology and its effect on the environmental impact of air transport

- the effect of government policies and global agreements that might affect the need to replace older aircraft by new aircraft that have a less damaging impact on the environment
- the efficiency (eg cost per seat) of operating the company's aircraft compared to its competitors
- the environmental impact of the company's aircraft compared to its competitors
- the safety record of the company's aircraft compared to its competitors
- the impact of unexpected events such as the 2007-08 financial crisis and the global pandemic
- consumer confidence in air travel
- the growth in the demand for air freight, perhaps linked to expected changes in the pattern of world trade
- use of extracts A, B and C to support the above.

The use of relevant diagrams to support analysis should be taken into account when assessing the quality of a candidate's response to the question.

[15]

23.

A

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24.

Areas for discussion include:

- explanation of the gig economy and the difference between terms of employment in the gig economy and standard employment
- discussion of the emerging trends that have contributed to the rise of the gig economy, such as technological progress and online commerce
- what is meant by the 'interests of people who are working in the gig economy'
- discussion of the reasons why some roles/industries are more likely to arise in the gig economy than others
- the benefits of gig economy work to the individual, such as: flexibility over hours, ease of switching between jobs, autonomy from having no supervising manager
- the costs of gig economy work to the individual, such as lower pay and fewer non-wage benefits (for example pension, health insurance, maternity benefits)
- analysis of why wages may be lower and working conditions worse in the gig economy than in the formal sector
- benefits and costs of the gig economy to firms and the wider economy, such as greater labour mobility, lower costs of production, economic efficiency
- discussion of the impact of the pandemic on the gig economy
- discussion of possible policies, such as legislation, raising minimum wages, guaranteeing minimum hours for all roles, redefining gig economy workers as 'employed' rather than 'self-employed'
- the argument that the need for government intervention is reduced by the existence and growth of trade unions representing gig economy workers
- the argument that the need for government intervention is reduced by the fact that some firms in the gig economy are improving conditions for their workers anyway
- use of examples
- market failure arguments
- government failure arguments.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the student's response to the question.

Level of response	Response	Max 25 marks
5	Sound, focused analysis and well-supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response - includes well-focused analysis with clear, logical chains of reasoning - includes supported evaluation throughout the response and in a final conclusion.	21–25 marks
4	Sound, focused analysis and some supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response - includes some well-focused analysis with clear, logical chains of reasoning - includes some reasonable, supported evaluation.	16–20 marks
3	Some reasonable analysis but generally unsupported evaluation that: - focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present - includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response - includes some reasonable analysis but which might not be adequately developed or becomes confused in places - includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgments but these aren't well-supported by arguments and/or data.	11–15 marks
2	A fairly weak response with some understanding that: - includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - includes some limited application of relevant economic principles to the given context and/or data to the question - includes some limited analysis but it may lack focus and/or become confused - includes attempted evaluation which is weak and unsupported.	6–10 marks
1	A very weak response that: - includes little relevant knowledge and understanding of economic terminology, concepts and principles - includes application to the given context which is, at best, very weak - includes attempted analysis which is weak and unsupported.	1–5 marks

[25]

25.

C

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